Sorting Algorithms - DSA Notes

Bubble Sort

What is it?
Compare adjacent elements and swap if needed. Largest element bubbles to the end after each pass.

Dry Run:
5 3 8 4 2 → 3 5 4 2 8 → 3 4 2 5 8 → 3 2 4 5 8 → 2 3 4 5 8

Complexity:
Best O(n), Avg O(n^2), Worst O(n^2), Space O(1)

Selection Sort

What is it?
Find the minimum element and place it at the correct position.

Dry Run:
5 3 8 4 2 → 2 3 8 4 5 → 2 3 4 8 5 → 2 3 4 5 8

Complexity:
O(n^2), Space O(1)

Insertion Sort

What is it?
Insert each element into the correct position in the sorted part.

Dry Run:
5 3 8 4 2 → 3 5 8 4 2 → 3 4 5 8 2 → 2 3 4 5 8

Complexity:
Best O(n), Avg/Worst O(n^2), Space O(1)

Merge Sort

What is it?
Divide array, sort halves, then merge.

Dry Run:
8 3 5 1 → [8 3][5 1] → [8][3][5][1] → [3 8][1 5] → [1 3 5 8]

Complexity:
O(n log n), Space O(n)

Quick Sort

What is it?
Choose a pivot and partition around it.

Dry Run:
4 7 2 9 1 (pivot=4) → 2 1 | 4 | 7 9 → 1 2 4 7 9

Complexity:
Avg O(n log n), Worst O(n^2), Space O(log n)

Heap Sort

What is it?
Build heap and repeatedly remove largest element.

Dry Run:
Build Max Heap → Extract largest repeatedly

Complexity:
O(n log n), Space O(1)

Counting Sort

What is it?
Count frequencies and rebuild array.

Dry Run:
4 2 2 1 → count → 1 2 2 4

Complexity:
O(n+k), Space O(k)

Radix Sort

What is it?
Sort digit by digit.

Dry Run:
170 45 75 90 → units → tens → hundreds

Complexity:
O(d*n), Space O(n)